**Paper 1: Determinants of Cattle Finishing Profitability**

Using ten years of monthly closeout data from a western Kansas feedlot, Langemeier, Schroeder, and Mintert (1992) found that approximately 98% of profit variability could be explained by six variables: fed cattle sale price, feeder cattle purchase price, corn price, interest rates, feed conversion ratio, and average daily gain. The study also showed that placement weight affects which variables matter most — for example, corn prices have a relatively larger impact on lighter-weight placements. The authors used standardized regression coefficients to rank each factor's contribution, giving feedlot managers a clear priority list for risk management. This remains one of the most-cited foundational papers in feedlot economics.

**Connection to FEDVT:** This paper directly justifies the cost category structure of the FEDVT dashboard. The six variables that explain 98% of profit variability are precisely the inputs your tool models. Cite this in Section 2 (Background) when explaining why those specific parameters were selected.

Citation: Langemeier, M. R., Schroeder, T. C., & Mintert, J. (1992). Determinants of cattle finishing profitability. *Journal of Agricultural and Applied Economics, 24*(2), 41–47. https://www.cambridge.org/core/journals/journal-of-agricultural-and-applied-economics/article/abs/determinants-of-cattle-finishing-profitability/3D4AE31249515FA5FDE3DA5931D78AD1

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**Paper 2: Elements of Cattle Feeding Profitability in Midwest Feedlots**

Lawrence, Wang, and Loy (1999) analyzed cross-sectional and time-series data from over 1,600 pens across more than 220 Midwest feedlots to examine what drives profitability beyond the standard price variables. In addition to input and output prices and animal performance, the study found that sex, placement weight, facility design, and placement season all significantly affect net returns. The findings challenged the prevailing assumption that prices alone determine feedlot margins, showing that operational and biological factors introduce meaningful variation even within similar market environments.

**Connection to FEDVT:** This paper supports the breadth of FEDVT's 65-parameter input structure. It justifies why the tool includes variables beyond just feed and cattle prices — sex, placement weight, and facility-level costs are all modeled in the dashboard because peer-reviewed literature confirms they matter. Cite in Section 3 (Materials & Methods) when explaining parameter selection.

Citation: Lawrence, J. D., Wang, Z., & Loy, D. (1999). Elements of cattle feeding profitability in Midwest feedlots. *Journal of Agricultural and Applied Economics, 31*(2), 349–357. https://www.cambridge.org/core/journals/journal-of-agricultural-and-applied-economics/article/abs/elements-of-cattle-feeding-profitability-in-midwest-feedlots/3B7F4D2BC5C7BAB49BA9DE941FD27F73

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**Paper 3: Identifying Economic Risk in Cattle Feeding**

Mark, Schroeder, and Jones (2000) used closeout data from two western Kansas commercial feedlots to measure how fed cattle prices, feeder prices, feed costs, and animal performance each contribute to profit variability. They applied standardized beta coefficients to compare these factors across sex, placement weight class, and placement season. Results showed that price risk dominates overall, but performance-related risk is proportionally larger for fall placements and lighter-weight cattle. The study concluded that fed and feeder cattle prices should receive the most emphasis in risk management strategies, though operational factors remain meaningful under specific placement conditions.

**Connection to FEDVT:** This is the primary justification for FEDVT's sensitivity analysis outputs. The finding that different variables dominate risk across different placement scenarios is exactly what a scenario simulation tool should demonstrate. Cite in Section 2 (Background) and Section 4 (Results) when discussing the sensitivity analysis visualization.

Citation: Mark, D. R., Schroeder, T. C., & Jones, R. (2000). Identifying economic risk in cattle feeding. *Journal of Agribusiness, 18*(3), 331–344. https://ageconsearch.umn.edu/record/14712

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**Paper 4: Management of Multiple Sources of Risk in Livestock Production**

McKendree, Tonsor, and Schulz (2021) surveyed feedlot operators using a split-sample choice experiment to understand how they manage output price risk and animal health risk jointly. The study found that operators who source cattle from a single known ranch are less likely to rely exclusively on spot markets, suggesting that risk management strategies across different domains are complementary rather than independent. Operators who reduce production risk through sourcing decisions tend to also engage more actively in price risk management. The study provided rare empirical data on actual feedlot risk management behavior rather than theoretical prescriptions.

**Connection to FEDVT:** Supports the Discussion section on stakeholder utility. Real feedlot operators manage multiple risk dimensions simultaneously — this paper validates FEDVT's multi-scenario, multi-variable design as aligned with how producers actually think and operate. Cite in Section 5 (Discussion) when addressing practical decision-making utility.

Citation: McKendree, M. G. S., Tonsor, G. T., & Schulz, L. L. (2021). Management of multiple sources of risk in livestock production. *Journal of Agricultural and Applied Economics, 53*(1), 75–93.
https://www.cambridge.org/core/journals/journal-of-agricultural-and-applied-economics/article/management-of-multiple-sources-of-risk-in-livestock-production/62D01C72A30AD7F559AF1F52D1BDD1A2

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**Paper 5: Multivariate Forecasting of a Commodity Portfolio: Application to Cattle Feeding Margins and Risk**

Tonsor and Schroeder (2011) argue that feedlot investors and managers operate in an inherently multivariate price environment — simultaneously exposed to live cattle, feeder cattle, and corn price risk — yet most decision tools model these prices as independent. The study develops multivariate simulation techniques that model the joint distribution of all three prices and evaluates how this approach improves forecasting accuracy for future feeding margins. Results show that properly accounting for the correlations among these prices significantly outperforms univariate approaches in predicting actual cash market outcomes. The authors conclude that decision tools for feedlot managers should reflect this multivariate price reality rather than treating each commodity in isolation.

**Connection to FEDVT:** This paper directly motivates FEDVT's future development path. The tool currently takes prices as user-specified inputs without modeling their joint behavior. Tonsor and Schroeder demonstrate what the next generation of the tool should do — simulate correlated price scenarios across all three commodity inputs simultaneously. Cite in Section 6 (Conclusion and Future Work) when discussing how FEDVT's scenario simulation capability could be extended.

Citation: Tonsor, G. T., & Schroeder, T. C. (2011). Multivariate forecasting of a commodity portfolio: application to cattle feeding margins and risk. *Applied Economics, 43*(11), 1329–1339. https://doi.org/10.1080/00036840802600517

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**Paper 6: Feed Prices, Substitution Patterns, and Technical Efficiency in Feedlot Cattle**

Dennis and Schroeder (2023) used Kansas Focus on Feedlots closeout data from January 2015 through December 2023 to examine how changes in feed prices affect not only direct feed costs but also feedlot technical efficiency via substitution effects. When corn prices rise, feedlots substitute toward alternative feedstuffs, which changes feed conversion ratios, average daily gain, and days on feed. The study found that net returns over the period ranged from +$457 to -$489 per head, illustrating the extreme margin volatility that feedlot operators face. Feed prices were identified as a key driver of both cost structure and biological performance simultaneously.

**Connection to FEDVT:** This is your most current data citation and one of the strongest justifications for FEDVT's existence. The ±$489 per-head return range is a powerful statistic for your Introduction to establish the economic stakes. Also validates the feed cost sensitivity scenarios in Section 4 with real recent data. Cite in Section 1 (Introduction) and Section 4 (Results).

Citation: Dennis, E. J., & Schroeder, T. C. (2023). Feed prices, substitution patterns, and technical efficiency in feedlot cattle. *Journal of Agricultural and Applied Economics, 55*(4), 546–566.
https://www.cambridge.org/core/journals/journal-of-agricultural-and-applied-economics/article/feed-prices-substitution-patterns-and-technical-efficiency-in-feedlot-cattle/610E2D9B123E242AF9F626AAF282C88E

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**Paper 7: Recent Advances in Decision Support for Beef and Dairy Farming**

Bang, Laugen, and Dreyer (2023) conducted a systematic review of 110 published articles on decision support systems for beef and dairy farming from 2016 to 2022. They classified these studies by farm type, modeling technique, and decision problem addressed. The review found that while academically sophisticated decision models are abundant, practically adopted, user-friendly tools that work in real farm settings remain scarce. The gap between modeling complexity and on-farm usability was identified as the most persistent problem in the field, with most tools never moving beyond proof-of-concept.

**Connection to FEDVT:** This is your single most important positioning citation. It provides comprehensive recent evidence confirming the exact gap FEDVT addresses — decision support tools exist but aren't reaching producers. Cite prominently in Section 2 (Background and Literature Review) to establish the knowledge gap your paper fills.

Citation: Bang, A. L., Laugen, A. T., & Dreyer, H. C. (2023). Recent advances in decision support for beef and dairy farming: modeling approaches and opportunities. *International Transactions in Operational Research, 30*(5), 2214–2244. https://doi.org/10.1111/itor.13311

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**Paper 8: Feeder Cattle Basis Risk and Determinants**

Bina and Schroeder (2022) used weekly auction transaction data from 32 feeder cattle markets across the U.S. from 1992 to 2021 to analyze the determinants of feeder cattle basis risk. The study found that feeder cattle market volatility has a statistically and economically significant influence on basis risk — the more volatile the cash market, the harder it is to hedge effectively. Basis risk peaked historically during the 2014–2015 cattle market disruption but had declined by 2018, though it remained elevated compared to pre-2014 levels. Lot characteristics, seasonal timing, and regional market conditions all contributed to basis variation.

**Connection to FEDVT:** Supports your Background section on why feedlot operators need producer-specific tools rather than generic market signals. Because basis varies significantly by location and lot characteristics, a tool that starts from the individual operation's own cost structure — like FEDVT — is more useful than relying on national-level futures data alone. Cite in Section 2 (Background).

Citation: Bina, J. D., & Schroeder, T. C. (2022). Feeder cattle basis risk and determinants. *Journal of Agricultural and Applied Economics, 54*(1), 1–21. https://www.cambridge.org/core/journals/journal-of-agricultural-and-applied-economics/article/feeder-cattle-basis-risk-and-determinants/5572C2B63BBB2ADAE442081F07F0C696

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**Paper 9: Economic Assessments from Experimental Research Trials of Feedlot Cattle Health and Performance: A Scoping Review**

Dixon, Hanthorn, Pendell, Cernicchiaro, and Renter (2022) conducted a scoping review of 113 feedlot trials from 91 peer-reviewed articles to evaluate how economic assessments are conducted and reported in feedlot cattle research. Despite the central importance of economics to producer decision-making, the study found substantial inconsistency — only 28% of trials fully reported economic outcomes, methodology, price values, and data sources together. Eight types of economic assessments were identified across the literature, including break-even analysis, partial budgeting, enterprise analysis, and profitability modeling. The authors concluded that without standardized frameworks for economic reporting, feedlot research results are difficult to compare across studies and less actionable for stakeholders.

**Connection to FEDVT:** This paper directly legitimizes FEDVT's design approach. The finding that break-even analysis and partial budgeting are the most practically relevant economic frameworks in feedlot research validates FEDVT's deterministic break-even calculator as methodologically grounded. It also supports your Discussion section's argument that a transparent, standardized tool improves on the fragmented, inconsistently reported economic analyses currently prevalent in the literature. Cite in Section 2 (Background) and Section 5 (Discussion) when arguing for FEDVT's contribution to economic transparency and operator accessibility.

Citation: Dixon, A. L., Hanthorn, C. J., Pendell, D. L., Cernicchiaro, N., & Renter, D. G. (2022). Economic assessments from experimental research trials of feedlot cattle health and performance: a scoping review. *Translational Animal Science, 6*(3), txac077. https://doi.org/10.1093/tas/txac077

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**Paper 10: Impacts of Changes in Market Fundamentals and Price Momentum on Hedging Live Cattle**

Coffey, Tonsor, and Schroeder (2018) analyzed basis prediction errors for live cattle across five major Mandatory Livestock Price Reporting regions to understand what makes hedging difficult. They found that cost-of-gain volatility and delivery cost variation are larger contributors to hedging difficulty than market trends or price momentum. Results differed significantly across regions, indicating that local market conditions matter more than national signals when designing a risk management strategy.

**Connection to FEDVT:** Reinforces the case for FEDVT as an individual-operation tool. Since hedging difficulty is driven primarily by local cost-of-gain volatility rather than national trends, operators need a tool grounded in their own cost structure — exactly what FEDVT provides through its customizable input parameters. Cite in Section 2 (Background) alongside source 9.

Citation: Coffey, B. K., Tonsor, G. T., & Schroeder, T. C. (2018). Impacts of changes in market fundamentals and price momentum on hedging live cattle. *Journal of Agricultural and Resource Economics, 43*(1), 18–33. https://doi.org/10.1017/aae.2018.31

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**Paper 11: An Evaluation of the Economic Effects of BRD on Feedlot Cattle**

Blakebrough-Hall, McMeniman, and González (2020) tracked 898 crossbred steers through an Australian feedlot and quantified BRD's economic impact using four diagnostic methods. The study found an 18% morbidity rate and a 2.1% BRD mortality rate. Animals treated three or more times produced 39.6 kg lighter carcasses and returned AUD $385 less per head than healthy animals. Even subclinical BRD — animals with lung lesions at slaughter but no treatment records — reduced net returns by AUD $67 per head. The findings demonstrated that BRD imposes substantial economic costs that are often invisible to standard pen-level management tracking.

**Connection to FEDVT:** Your primary BRD economics citation for the disease outbreak use-case in Section 4 (Results). These figures give you peer-reviewed benchmarks for realistic health cost inputs in FEDVT's scenario simulation. Cite when demonstrating the disease outbreak scenario to show the tool's outputs are calibrated against real-world data.

Citation: Blakebrough-Hall, C., McMeniman, J. P., & González, L. A. (2020). An evaluation of the economic effects of bovine respiratory disease on animal performance, carcass traits, and economic outcomes in feedlot cattle defined using four BRD diagnosis methods. *Journal of Animal Science, 98*(2), skaa045. https://doi.org/10.1093/jas/skaa045

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**Paper 12: Market Impacts of Reducing the Prevalence of BRD in US Feedlots**

Karle, Toth, and White (2017) used a multi-market partial equilibrium model of the U.S. agricultural industry to simulate the market-wide effects of reducing BRD prevalence in feedlots. Using NAHMS Beef Feedlot 2011 data showing 21.2% of placed cattle were affected by BRD at a direct treatment cost of $23.60 per case, they estimated the supply and demand ripple effects of a hypothetical BRD reduction. They found that while individual producers benefit from reduced morbidity and mortality, market-level price adjustments partially offset those gains. Consumer welfare increased overall due to lower beef prices resulting from higher supply.

**Connection to FEDVT:** Provides the industry-scale context for BRD's economic significance in your Background section. The $23.60 per-case figure and 21.2% morbidity rate are widely cited benchmarks you can use when parameterizing the health cost inputs in FEDVT's scenario demonstrations. Cite in Section 2 (Background).

Citation: Karle, A. S., Toth, J. D., & White, B. J. (2017). Market impacts of reducing the prevalence of bovine respiratory disease in United States beef cattle feedlots. *Frontiers in Veterinary Science, 4*, 189. https://doi.org/10.3389/fvets.2017.00189

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**Paper 13: Structural Changes in Feedlot Performance Over Time**

Herrington and Tonsor (2013) used the Kansas Focus on Feedlots dataset spanning multiple decades to document long-run structural changes in feedlot performance. They found that average daily gain and feed conversion efficiency improved over time, reflecting genetic and nutritional advances, but that margin volatility did not decrease correspondingly. The feedlot operating environment grew more complex over the study period, with performance improving but financial risk not diminishing. The study highlighted that production improvements alone do not insulate operators from economic uncertainty.

**Connection to FEDVT:** This is your knowledge gap citation. Feedlots have become more technically sophisticated, yet the tools available to operators — primarily static spreadsheets — have not kept pace. FEDVT's interactive, scenario-capable design responds directly to the dynamic complexity documented here. Cite in Section 2 (Background) when establishing the gap between operational complexity and available decision tools.

Citation: Herrington, M. A., & Tonsor, G. T. (2013). Econometric estimations of performance improvements in Kansas feedlot cattle. *The Professional Animal Scientist, 29*(4), 435–442. https://www.appliedanimalscience.org/article/S1080-7446(15)30257-6/fulltext

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**Paper 14: Price-Weight Relationships for Feeder Cattle**

Dhuyvetter and Schroeder (2000) examined price-weight slide relationships for feeder cattle using Canadian auction market data, demonstrating that the price per pound paid declines as animal weight increases and that this relationship shifts with corn prices and overall cattle market conditions. They quantified how much the price slide steepens during periods of high feed costs, since heavier cattle require more feed to finish and carry greater cost risk. The study provided empirically estimated slide coefficients that can be used directly in budget and break-even calculations.

**Connection to FEDVT:** Directly relevant to FEDVT's feeder cattle purchase cost calculations, which represent the dominant cost driver in your tool. This paper provides the economic mechanism behind the price-weight relationship that operators must account for when entering feeder cattle costs. Cite in Section 3 (Materials and Methods) when explaining the feeder cattle cost parameterization.

Citation: Dhuyvetter, K. C., & Schroeder, T. C. (2000). Price-weight relationships for feeder cattle. *Canadian Journal of Agricultural Economics, 48*(3), 299–310. https://doi.org/10.1111/j.1744-7976.2000.tb00282.x

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**Paper 15: Simulation Approaches Used for Management and Decision Making in the Beef Production Sector: A Systematic Review**

Awasthi et al. (2024) reviewed 105 simulation modeling studies in beef production published through June 2023, classifying them by modeling approach, focus area, and validation method. The review found that most models are deterministic and dynamic, with fewer than half including formal validation procedures. Excel, R, Python, and STELLA were the most commonly used tools. The authors identified inconsistent reporting standards and limited transparency as the main barriers to reproducibility and real-world adoption of these models. They concluded that standardized development and validation protocols are needed to make simulation tools more credible and useful to practitioners.

**Connection to FEDVT:** Provides a broad systems-level justification for simulation-based decision tools in beef production and highlights the exact methodological gaps FEDVT should address — particularly the need for transparent documentation and validation. Cite in Section 2 (Background) and use the call for standardized tools to strengthen your argument for FEDVT's structured design.

Citation: Awasthi, T. R., Morshed, A., Williams, T., & Swain, D. L. (2024). Simulation approaches used for management and decision making in the beef production sector: A systematic review. *Animals, 14*(11), 1632. https://doi.org/10.3390/ani14111632

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**Paper 16: Improvement of Feedlot Operations Through Statistical Learning and Business Analytics Tools**

Flores et al. (2017) proposed using statistical learning and business analytics to improve feedlot decision-making, particularly around the timing of cattle marketing. The study introduced a decision-support tool that predicts individual animal growth and profitability using measurable traits such as ultrasound backfat, marbling scores, and feeding performance data. By modeling these factors, the tool estimates the optimal slaughter date for each animal to maximize profit and minimize unnecessary feed costs. The authors showed that even with limited data, predictive models can meaningfully improve feedlot efficiency, while also acknowledging barriers related to data quality, market variability, and industry adoption.

**Connection to FEDVT:** A strong precedent citation showing that analytics-based decision support tools for feedlot management have prior peer-reviewed validation. Positions FEDVT within an emerging class of data-driven tools, while also contrasting with it — Flores et al. focus on individual animal optimization, while FEDVT operates at the enterprise level. Cite in Section 2 (Background) to establish the broader analytics-in-feedlots context.

Citation: Flores, H., Meneses, C., Villalobos, J. R., & Sanchez, O. (2017). Improvement of feedlot operations through statistical learning and business analytics tools. *Computers and Electronics in Agriculture, 143*, 273–285. https://www.sciencedirect.com/science/article/abs/pii/S0168169916312030

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**Paper 17: Improving Feedlot Profitability Using Operational Data in Mortality Prediction Modeling**

Feuz, Feuz, and Johnson (2021) used machine learning models — logistic regression, decision trees, random forest, k-nearest neighbor, and naïve Bayes — trained on data from over 4,400 cattle to predict mortality and assist managers in making objective culling decisions. Logistic regression performed best at approximately 95.6% accuracy. A simulated economic analysis found that using the model could increase average net returns by $14.01 per head for previously treated cattle, and applying cost-sensitive learning pushed that figure to $45.27 per head with less than 1% probability of a loss. The study concluded that predictive analytics can substantially improve profitability at scale in large commercial feedlots.

**Connection to FEDVT:** Directly relevant to FEDVT's health and veterinary cost modeling and the disease scenario use-case in Section 4. This paper demonstrates the financial magnitude of data-driven health decisions and provides concrete per-head dollar figures that contextualize FEDVT's health cost inputs. Cite in Section 2 (Background) and Section 4 (Results) when presenting disease outbreak scenarios.

Citation: Feuz, R., Feuz, K., & Johnson, M. (2021). Improving feedlot profitability using operational data in mortality prediction modeling. *Journal of Agricultural and Resource Economics, 46*(2), 242–255. https://www.jstor.org/stable/27154110

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**Paper 18: Optimizing Feedlot Placement Weights Using Simulation-Based Mathematical Programming**

Stika (2025) developed a mathematical programming model to identify the optimal feeder cattle placement weight for maximizing net returns in commercial feedlots. Using real-world Kansas feedlot data and Python-based simulation, the model found that the optimal placement weight is approximately 704 pounds with a feeding duration of 166 days, producing net returns of around

$119 per head. The profit function incorporated all major variable costs — feeder purchase, feed, yardage, health, and interest — along with biological performance parameters. Sensitivity analysis showed that feeder cattle prices and feed costs have the largest impact on profitability, while moderate changes in corn price or dressing percentage have smaller effects.

**Connection to FEDVT:** Directly parallel to FEDVT's design objectives. Both tools aim to translate feedlot cost structure into actionable placement and marketing decisions. Stika's finding that feeder price and feed cost dominate sensitivity results independently validates FEDVT's emphasis on those same parameters. Cite in Section 2 (Background) as a recent peer-reviewed counterpart to your tool's approach.

Citation: Stika, M. N. (2025). Optimizing feedlot placement weights using simulation-based mathematical programming. Kansas State University. https://krex.k-state.edu/items/660535d7-67d3-433f-be1b-28d88d044c62

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**Paper 19: Using Probabilistic Machine Learning Methods to Improve Beef Cattle Price Modeling**

Rahmani, Khatami, and Stephens (2024) compared ARIMA, SARIMA, and SARIMAX time series models against multivariate machine learning methods — support vector regression, random forest, and AdaBoost — for forecasting fed steer cattle prices in Alberta, Canada using monthly data from 2005 to 2023. Random forest and AdaBoost outperformed the univariate time series models. The authors then added a probabilistic modeling layer on top of random forest to generate prediction intervals rather than point estimates, allowing producers to see the uncertainty range around any forecast. They acknowledged limitations in capturing extreme market events and recommended future work with ensemble approaches and additional economic predictors.

**Connection to FEDVT:** Highly relevant to Section 2 (Background) as it validates the broader forecasting methodology used in your project's ML pipeline while also directly supporting the Future Work section of your paper. The recommendation for probabilistic forecasting intervals mirrors FEDVT's future direction as described in Belasco et al. Cite in Background when discussing price forecasting context and in Future Work when discussing model uncertainty.

Citation: Rahmani, E., Khatami, M., & Stephens, E. (2024). Using probabilistic machine learning methods to improve beef cattle price modeling and promote beef production efficiency and sustainability in Canada. *Sustainability, 16*(5), 1789. https://doi.org/10.3390/su16051789

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**Paper 20: Application of Models in Feedlot Systems: Maintenance Energy Requirements, Growth and Cost Curves, and Feeding Behavior**

Harrison (2022) applied the Davis Growth Model to data from 120 Angus-cross steers fed through both automated feeding systems and conventional bunks to predict maintenance energy requirements, growth trajectories, and economic outcomes. The study found that modern cattle have higher maintenance energy and protein synthesis requirements than historical model estimates suggest. It also developed profit and cost curves to determine optimal harvest timing, finding that sorting cattle by predicted days on feed rather than body weight improved carcass uniformity and reduced overfeeding costs. The research demonstrated that integrating mechanistic growth models with precision livestock technologies can move feedlots from pen-level to individualized management.

**Connection to FEDVT:** Supports both the Background and Future Work sections. Harrison's use of cost curves and profit optimization as a management output is conceptually aligned with FEDVT's break-even and profitability projection outputs. The call for individualized, data-driven feedlot management also connects to FEDVT's future integration potential with IoT and precision livestock data, as described in Section 5 (Discussion). Cite in both Section 2 and Section 5.

Citation: Harrison, M. A. (2022). *Application of models in feedlot systems: Maintenance energy requirements, growth and cost curves, and feeding behavior* (Order No. 29995958). ProQuest Dissertations & Theses Global.
http://proxy.library.tamu.edu/login?url=https://www.proquest.com/dissertations-theses/application-models-feedlot-systems-maintenance/docview/2788148791/se-2